Research article

Growth Requirements for High-Redshift Black Holes: A JWST-Era Catalogue

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Abstract

Spectroscopy with the James Webb Space Telescope (JWST) pushes black-hole mass measurements into the early Universe, when black holes had less time to grow. We test which combinations of seed mass, accretion rate, starting time, radiative efficiency, and merger contribution can reproduce these masses within the available time. We present growth requirements and parameter maps for 238 black holes and candidates studied with JWST at $z \geq 4$, spanning 18,207,000 grid evaluations across seed mass, accretion rate, and growth-onset assumptions. Required accretion rates, seed masses, and duty cycles are linked to each object's adopted published measurement. The primary sample contains 223 objects with broad Balmer-line mass estimates; the expanded sample includes tentative or conditional candidates and GN-z11. We propagate reported mass errors with Monte Carlo sampling and test systematic mass shifts and alternative measurements. For $100\,M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no mergers, eleven primary objects require lifetime-average Eddington ratios above unity. Eight exceed unity in at least 95% of mass draws; five retain this threshold after reducing masses by 0.5 dex. Adding separately reported calibration scatter lowers two of these eight probabilities below 95%. Reducing only the efficiency to $\epsilon = 0.05719$, the ideal thin-disc efficiency for zero spin, allows all 238 central mass estimates to be reached without lifetime-average accretion above the Eddington limit. Independent mass measurements would test whether the highest growth requirements survive systematic mass revisions.

Keywords: Supermassive black holes, Accretion, Active galactic nuclei, High-redshift galaxies, Astronomical catalogs

1 Introduction

The masses and redshifts of early black holes constrain the growth needed from an assumed seed. JWST surveys supply broad-line mass estimates across different selection methods and observing fields (Harikane et al. 2023; Matthee et al. 2024; Baccus & Xu 2026). Growth studies of individual objects, including UNCOVER-20466 and GN-z11, compare light seeds, heavy seeds, and rapid accretion (Kokorev et al. 2023; Maiolino et al. 2024).

Comparing growth histories across surveys requires tracking which measurements refer to the same object, which mass estimate is adopted, and which alternatives remain plausible. Tentative line detections and differences in calibration coverage also affect which objects can enter a common comparison.

We compile adopted and alternative measurements across JWST surveys and apply a common growth calculation to identify which objects remain demanding under mass revisions and changes in growth assumptions. We release accretion-rate, seed-mass, and duty-cycle requirements linked to the adopted measurements and source versions, together with uncertainty summaries and parameter maps. The compilation contains 355 measurements for 343 provisional objects. Growth calculations use 238 objects with suitable masses after excluding unresolved object matches, including 223 primary objects with broad Balmer-line mass estimates. Section 3.4 describes the released constraints.

Using the exponential growth relation applied by Dayal (2024), we rank required lifetime-average Eddington ratios and test their sensitivity to reported mass errors, mass revisions, seed mass, growth onset, and radiative efficiency. A separate test adds quantified calibration scatter for three objects. A four-target extension tests earlier accretion onset (Appendix E).

2 Measurement catalogue

2.1 Scope and source coverage

We compile 355 literature measurements from 37 source families (studies or related surveys) at $z \geq 4$, with a literature cutoff of 3 September 2026. We admit a study when a stable publication or archive version provides extractable measurements, sufficient information to assess lensing and object matches, and documented mass methods and uncertainty conventions where masses are reported. Adopted preprint versions and later published alternatives are recorded separately. The source inventory defines the compilation; it is not an exhaustive literature census and does not systematically include legacy ground-selected quasars. These records represent 343 provisional objects and 342 host systems. Table 1 summarizes the catalogue and analysis sample sizes. Figure 1 shows the sample accounting; Table 2 lists the contributing source families.

Table 1 here.

Figure 1 here.

Table 2 here.

2.2 Adopted measurements and analysis samples

We adopt one measurement per object and retain alternatives, recording the source, method, uncertainty, publication version, and reason for each choice (Appendix A.3).

Growth analysis requires a numerical mass and redshift, an identified mass method, and a documented lensing correction where needed. Upper limits, indirect model ranges, and masses inferred by assuming an Eddington ratio are excluded from numerical inference. Entries with unresolved identities across studies are also excluded (Appendix A.3).

For the primary sample, we use masses estimated from broad Hα or Hβ emission lines in objects with secure or probable evidence for an accreting black hole, using the evidence classifications recorded from the source papers. Candidates with tentative broad-line detections or masses conditional on a broad-line-region interpretation enter the expanded sample only. Evidence for an active galactic nucleus (AGN) or a broad-line region determines sample membership; potential biases in an accepted virial mass are assessed separately. COSMOS3D-13852 therefore remains in the primary sample on the basis of its published broad-line AGN identification; we test possible scattering bias in Section 5.2.

These single-epoch virial estimators combine the line width, which estimates gas velocity, with a luminosity-based estimate of the broad-line region's radius. Their calibration is based on reverberation mapping of nearby AGN (Shen 2013). Using the same line family and mass-estimation framework reduces differences between methods in the comparison. Applying these calibrations at high redshift still requires assessing gas dynamics, geometry, obscuration, and lensing for each object.

The expanded sample adds 7 candidates with masses conditional on a broad-line-region interpretation, 7 with tentative broad-line detections, and GN-z11 to the primary sample (Appendix A.2). GN-z11's mass is inferred from rest-frame ultraviolet emission lines using local virial relations (Maiolino et al. 2024); its lines and calibration differ from those of the Balmer-line sample. These criteria yield 223 primary objects and 238 expanded-sample objects.

Objects without reported mass errors enter calculations at their central masses but not the uncertainty analysis. The mass estimates retained in each sample still use different calibrations, and the contributing studies have different target-selection criteria and detection limits. Appendix A gives source-version notes, individual membership decisions, and measurement-choice checks.

2.3 Released measurement tables

The release contains measurement records, one adopted record per provisional object, and flags selecting the primary and expanded samples. Appendix A.5 gives the table locations, join keys, units, and missing-value conventions. Section 3.4 describes the growth products.

2.4 Validation

We check all 248 numerical mass records against reference values separately transcribed from the publications, including missing errors for 12 records. The reference values are not taken from the generated catalogue. Separate checks cover all 355 central redshifts and 328 available coordinate pairs. The records retain source versions, table locators, and conversion notes. These checks test transcription and unit

conversion; they do not validate the mass estimators or resolve ambiguous identities. Other fields, including luminosities, magnifications, and evidence classifications, have partial independent coverage. The fixtures and verification commands are in `data/validation/`.

2.4.1 AI assistance

OpenAI models `gpt-5.6-sol`, `gpt-6-astra`, and `gpt-6.1-sol` assisted with scraping published papers for catalogue data and with code review, repository documentation, code development, and manuscript drafting and editing through Codex. The author is responsible for the analysis, interpretation, and final manuscript.

3 Derivation of growth constraints

3.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$, where $L_{\rm bol}$ is the bolometric luminosity of the accretion flow and $L_{\rm Edd}$ is the Eddington luminosity for electron scattering in ionized hydrogen. The fixed radiative efficiency ϵ is the fraction of inflowing rest-mass energy emitted as radiation; the fraction $1-\epsilon$ is retained by the black hole in this model. We define the available growth interval Δt and the lifetime-average Eddington ratio $\overline{f}_{\rm Edd}$ as

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f}_{\rm Edd} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\,dt. \tag{1}$$

Here, $t(z)$ is the age of the Universe at redshift z, $z_{\rm seed}$ is the redshift at which the seed forms and growth begins, and $z_{\rm obs}$ is the observed redshift. The average includes active and inactive phases. We use the exponential growth relation adopted by Dayal (2024), expressed in terms of the time-averaged Eddington ratio, and introduce a multiplier $B_{\rm merge}$ to approximate additional mass from mergers:

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\rm Edd}\,\Delta t}{t_{\rm Edd}}\right], \tag{2}$$

Here, $M_{\rm BH}(t_{\rm obs})$ is the black-hole mass at the observation time $t_{\rm obs} = t(z_{\rm obs})$, and $M_{\rm seed}$ is the initial seed mass. The Eddington timescale is $t_{\rm Edd} = M_{\rm BH}c^2/L_{\rm Edd} \simeq 0.45$ Gyr, where c is the speed of light. The mass-growth e-folding time is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$ (Appendix B). The dimensionless merger multiplier is $B_{\rm merge} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. We represent mergers by multiplying the seed mass by a constant factor before calculating accretion. This approximation omits the timing and sequence of mergers, their mass ratios, and mass lost through gravitational radiation. The required lifetime-average ratio $\overline{f}_{\rm Edd,req}$ is the

average needed to reach the observed mass $M_{\rm BH}$ from the assumed seed within Δt:

$$\overline{f}_{\rm Edd,req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right]. \tag{3}$$

If $B_{\rm merge}M_{\rm seed}$ exceeds the observed mass, we set the required ratio to zero, but the seed and merger contribution already overpredict the mass. Such a history cannot reproduce the observation without mass loss. Inferring a duty cycle also requires assumptions about the rates during active and inactive phases.

We calculate cosmic ages in a flat cosmology with $H_0 = 67.3\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$, neglecting radiation. The first two values follow Dayal (2024). Appendix B gives the derivations, implementation, and inverse seed-mass relation. Equations 2–3 assume constant radiative efficiency. The ideal thin-disc efficiencies used in the illustrative tracks (Appendix B.6) are 0.03775, 0.05719, and 0.42265 for dimensionless spins $a = -1, 0, +1$, respectively (Bardeen et al. 1972). Negative and positive spins are counter-aligned and aligned with the disc, respectively. Each track holds efficiency constant; spin evolution is unspecified.

3.2 Reference model

We compare objects using Eq. 3 with $M_{\rm seed} = 100\,M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, and $B_{\rm merge} = 1$. The seed mass represents a light remnant of a metal-free (Population III) star, as considered by Dayal (2024). Starting growth at $z = 30$ gives light seeds a long growth interval; later onset requires higher average accretion rates for the same final mass. The efficiency $\epsilon = 0.1$ matches the analytic growth calculation in Dayal (2024). Holding it constant allows us to infer the required average rate without specifying the black hole's spin history. Setting $B_{\rm merge} = 1$ isolates the growth supplied by gas accretion.

These choices define a common light-seed benchmark for ranking objects by their required lifetime-average Eddington ratios. Section 4.3 tests larger seeds, later onset, and lower efficiency; the compatibility calculations also include merger contributions (Appendix C).

3.3 Uncertainty and sensitivity

We use Monte Carlo sampling to propagate reported asymmetric log-mass errors, with 10,000 draws per object. Each draw has equal probability of falling above or below the central mass, with its distance drawn from a half-normal distribution scaled by the corresponding upper or lower error. The central value is thus treated as a median. We use the quoted errors as one-sigma Gaussian scales, including when their confidence level is unspecified; we do not rescale them to a common confidence level. The adopted JADES records, including GN-38509 in Table 4, do not establish the confidence-level convention; MACS0416-Y1 also has an unspecified interval confidence level. This is an approximation to the reported intervals, not a reconstruction of the source posterior. Redshifts are held fixed. For each sampled mass, we calculate the required average Eddington ratio and report the fraction of draws above unity,

denoted P. Thus $P \geq 0.95$ means that at least 95% of draws exceed unity under this error model and the fixed growth assumptions. It is not a uniformly calibrated confidence statement across the sample. Quoted errors also differ in whether they include calibration scatter. Probability calculations use objects with reported mass uncertainties (Table 1).

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5, -0.3, 0, $+0.3$, and $+0.5$ dex, keeping the sample and growth assumptions fixed. These offsets test shifts comparable to the uncertainty of single-epoch mass estimates (Shen 2013). For each offset, we report the fraction of shifted Monte Carlo draws that require an average Eddington ratio above 1 under the reference model. Sampling and conversion details are given in Appendix B.5.

For RUBIES-EGS-55604 and J0910_2028_12910, we separately treat the reported 0.5-dex calibration scale as the standard deviation of an independent, zero-mean Gaussian error in log mass. Their reported errors are symmetric, so the combined standard deviation is $\sigma_{\rm tot} = \sqrt{\sigma^2_{\rm reported} + 0.5^2}$ dex. We evaluate the resulting normal tail above the mass required for $\overline{f}_{\rm Edd,req} = 1$ exactly, at offsets of 0 and -0.5 dex. For J1007_AGN, we convolve the asymmetric reported-error model above with independent Gaussian scatter of 0.43 dex (Schindler et al. 2025). We integrate the two half-normal contributions numerically (Appendix B.5). This assumption tests scatter, separately from a fixed mass bias. We add no scatter to GN-38509 or ZS7, whose reported errors already include it, and assign none to objects without a separately quantified scale. This is a three-object sensitivity test, not a common error model for the full sample.

3.4 Released growth constraints and parameter maps

Table 3 summarizes the growth products for the 238-object expanded sample, including the 223 primary objects. The four core tables give required accretion rates, seed masses, duty cycles, and checks of whether specified growth conditions reproduce each adopted mass. They contain 80 entries per object, or 19,040 in total. The rate, seed-mass, and duty-cycle tables use $z_{\rm seed} = 30$, $\epsilon = 0.1$, and no mergers; the compatibility checks vary efficiency, merger contribution, and accretion rate across four seed-mass intervals. In the compatibility calculations, efficiency depends on spin and accretion rate but remains constant within each tested history (Appendix C.1).

Table 3 here.

Two maps per object show how the growth requirement changes with seed mass and either accretion rate or starting redshift. The first varies spin-dependent efficiency and merger contribution at $z_{\rm seed} = 30$; the second fixes $\epsilon = 0.1$ and assumes no mergers. Together, the 476 maps contain 18,207,000 grid evaluations using central mass estimates.

Uncertainty calculations use 10,000 draws for each of 226 objects; the remaining 12 contribute central-mass results only. Mass-offset tests reuse the same draws. These counts exclude unresolved object matches and objects without suitable masses. Appendix A.5 gives file locations, parameter grids, selection flags, and additional table products. The four-target earlier-onset extension is reported separately in Appendix E.

4 Catalogue growth constraints

4.1 Required accretion under the reference assumptions

For a $10^2\,M_\odot$ seed under the reference model, the median required lifetime-average Eddington ratio across the 223 primary objects is $\overline{f}_{\rm Edd,req} = 0.574$ (interquartile range 0.487–0.679), with eleven objects above unity and none above 2. Across the expanded 238-object sample, the corresponding median is 0.581 (interquartile range 0.488–0.689), with 17 above unity. These statistics use the required ratios at the adopted central masses. CAPERS-LRD-z9 has the highest required ratio, 1.519, with a 16th–84th percentile interval of 1.479–1.561 from its mass-error draws. UNCOVER-20466 (1.452) and CANUCS-LRD-z8.6 (1.448) follow. Table 4 lists the eleven primary objects above the threshold, ranked by the required ratios at their central masses.

The shorter growth time at $z = 10.603$ places GN-z11 fourth in the expanded-sample required-rate ranking despite its smaller mass. Table 4 here.

4.2 Sensitivity to mass estimates

Of the eleven primary objects above unity at their central masses, nine remain above it at the 16th percentile and eight have $P \geq 0.95$ (Fig. 2). The calculation uses the 211 primary objects with reported mass errors; Table 4 records their calibration coverage. Appendix A.4 gives the expanded-sample percentile counts. Figure 2 here.

Downward mass shifts of 0.3 and 0.5 dex reduce the number of primary objects above unity at their central masses from eleven to nine and seven, respectively (Table 5). CAPERS-LRD-z9, CANUCS-LRD-z8.6, UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. Source-specific biases that may exceed this offset are discussed in Section 5.2. Table 5 here.

The separate calibration test changes the interpretation of J0910_2028_12910: its unshifted probability falls from above 0.999 to 0.745 (Table 6). RUBIES-EGS-55604 retains a probability of 0.979 after both the added scatter and the -0.5 dex shift. For J1007_AGN, adding its 0.43-dex calibration scale lowers the unshifted probability from 0.997 to 0.901; after the mass shift it is 0.607. Two of the eight reported-error threshold classifications therefore change when separately quantified calibration scatter is included. The Gaussian-scatter assumption and its limited coverage are specified in Section 3.3. Table 6 here.

Adopting the published Baccus measurements and varying the treatment of unmatched or uncertain identities leaves the threshold counts and five highest-ranked objects unchanged. Alternative measurements for six other objects also leave them below the reference threshold (Appendix A.4).

4.3 Seed mass, growth onset, and radiative efficiency

Increasing the seed mass to $10^3\,M_\odot$ reduces the number of primary objects with required ratios above unity at their central masses from 11 to 5; for $10^5\,M_\odot$ seeds, no objects in either sample exceed 1. Delaying seeding to $z = 20$ raises the primary count to 21. Adopting $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black

hole ($a = 0$; Bardeen et al. 1972), leaves no objects in either sample above unity at their central masses (Table 7). Each test changes one reference parameter. Table 7 here.

GN-z11 requires an average Eddington ratio of 1.437 under the reference assumptions. Delaying seeding to $z = 20$ raises this to 1.880, above UNCOVER-20466's 1.733, reversing their order. Appendix D shows the dependence on seed mass and starting redshift for both objects.

Figure 3 shows the maximum fixed efficiency allowing $\overline{f}_{\rm Edd,req} \leq 1$ as a function of seed mass for the five primary objects retaining $P \geq 0.95$ after a -0.5 dex mass shift and GN-z11. The $z_{\rm seed} = 20$ and 30 curves show GN-z11's greater sensitivity to the starting time. These boundaries use central mass estimates and assume no merger boost. Illustrative growth tracks are given in Appendix B.6. Figure 3 here.

Appendix C gives the fractions of objects whose masses can be reproduced from each seed-mass range under specified accretion rates, spin-dependent efficiencies, and merger contributions.

4.4 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the virial mass adopted in the catalogue $\log_{10}(M_{\rm BH}/M_\odot) = 7.3 \pm 0.2$ (Greene et al. 2024) with the independent MOKA3D dynamical mass estimate 7.7 ± 0.3 (Juodžbalis et al. 2026b), using $z = 7.04$ and the reference model. The dynamical estimate depends on lens reconstruction and inclination. Using symmetric normal approximations to the quoted log-mass errors, the required average Eddington ratio rises from 0.929 to approximately unity, with 16th–84th percentile intervals of 0.895–0.964 and 0.947–1.052, respectively. For the dynamical mass, $P(\overline{f}_{\rm Edd,req} > 1) = 0.497$; the small departure from unity depends on the cosmic-age approximation (Appendix B.3) and is negligible relative to the mass uncertainty. Calibrating the virial sample would require independent measurements for more objects.

5 Discussion

5.1 Implications for black-hole growth

Most primary objects have required average Eddington ratios below unity under the reference assumptions, permitting growth from $100\,M_\odot$ seeds without super-Eddington accretion. The required average includes inactive periods, so longer interruptions demand higher rates during active phases. For the eleven objects above unity, reaching the adopted mass requires super-Eddington phases or a change in seed mass, starting time, efficiency, merger contribution, or mass estimate.

Lower radiative efficiency allows more mass to be retained for the same radiated luminosity. At the fixed ideal thin-disc efficiency for zero spin, $\epsilon = 0.05719$, all adopted central masses can be reached from light seeds with average Eddington ratios below unity. This imposed efficiency does not model spin evolution. The feasibility of these histories depends on whether gas can sustain the required accretion over the available time. For UNCOVER-20466 and GS-20057765, the required averages are 0.793 and 0.740 for $100\,M_\odot$ seeds formed at $z = 30$ without mergers. If accretion occurs at

the Eddington limit during active phases and ceases between them, the objects must remain active for about 79% and 74% of that interval. GS-20057765's mass depends on its tentative broad-Hβ component tracing a broad-line region.

At $\epsilon = 0.1$, $z_{\rm seed} = 30$, and without mergers, UNCOVER-20466 and GS-20057765 require average ratios above unity for $10^3 M_\odot$ seeds, but below unity for $10^5 M_\odot$ seeds.

5.2 Mass measurements and follow-up priorities

The five primary objects retaining $P \geq 0.95$ after a -0.5 dex mass shift are priorities for independent mass measurements. Table 4 gives the mass reductions needed to bring their central required ratios to unity; Table 8 identifies the relevant measurement limitations.

Table 8 here.

Lin et al. (2026) discuss possible 1–2 dex virial-mass overestimates from electron scattering. Applying these shifts to COSMOS3D-13852 lowers its required average ratio to 1.115 and 0.916, respectively, under the reference assumptions. Its reduction to the threshold is 1.576 dex, within that range. These shifts test a possible bias whose magnitude remains unmeasured for this object. UNCOVER-20466 requires a reduction of 1.921 dex; independent checks of its lensing and broad-line interpretation would test that mass.

J0910_2028_12910 reaches the reference threshold after a reduction of 0.330 dex, smaller than its separately reported 0.5-dex calibration scale (Baccus & Xu 2026). Its 0.01-dex quoted error therefore gives a misleading impression of precision if considered alone. The separate-scatter test in Table 6 quantifies the effect of the calibration scale. Independent estimates should also be checked for shared calibration errors.

For CEERS-1019, confirming the weak broad Hβ component would strengthen the basis of its virial mass (Larson et al. 2023). For GN-z11, a Balmer-line mass estimate would test the result obtained from ultraviolet lines.

5.3 Limitations

The contributing studies have different selection criteria and detection limits. Inferring population fractions from these sample counts requires completeness corrections. Unresolved object matches also affect catalogue counts; the selection and measurement checks are given in Appendix A.4.

The reported mass errors have uneven calibration coverage, and geometry, lensing, and line-broadening biases remain incompletely quantified. The probabilities describe the adopted mass-error model at fixed growth parameters. Accounting for uncertainty in those parameters requires justified distributions and correlations.

Equation 2 uses constant radiative efficiency and a fixed merger multiplier. Gas supply, feedback, spin evolution, and individual accretion histories require additional modelling. Distinguishing seed origins further requires formation models and population information. The cosmological approximation is tested in Appendix B.3.

6 Conclusions

We release growth requirements and parameter maps linked to published measurements for 238 high-redshift black holes and candidates. The main findings are:

- Under the reference assumptions of 100 $M_\odot$ seeds formed at $z = 30$, $\epsilon = 0.1$, and no mergers, eleven of the 223 primary objects require average Eddington ratios above unity. Eight exceed unity in at least 95% of mass-error draws, and five retain this threshold after a 0.5-dex downward mass shift. These probabilities depend on the adopted errors and their calibration coverage.
- Adding separately quantified calibration scatter lowers the probabilities for J0910_2028_12910 and J1007_AGN to 0.745 and 0.901, respectively. Two of the eight reported-error threshold classifications therefore change. This test covers three objects; other mass-estimator systematics remain unquantified.
- Reducing only the efficiency to $\epsilon = 0.05719$ permits every expanded-sample central mass to be reached with an average Eddington ratio below unity. Larger seeds also reduce the required rates. At the reference efficiency $\epsilon = 0.1$, delaying onset to $z = 20$ makes GN-z11 the highest-ranked object, with a required average Eddington ratio of 1.880.

Growth models must explain how gas supply and accretion histories could sustain the required rates. Independent mass measurements of the highest-ranked objects would establish which requirements survive revisions to the observational inputs.

Data and code availability

Code and data are available at https://github.com/nniickels/highz-accretion-atlas. The three-object calibration sensitivity calculation is preserved at commit `f437e6bfec4b`. The measurement compilation and four core growth tables correspond to commit `a1bef516db5c`. The four additional tables described in Appendix A.5 are generated by `src/internal/catalogue_products.py`, with settings in `data/scenarios/catalogue_growth_comparisons.json` and input and output hashes in `data/manifests/v3-dataset-manifest.json`. Earlier-onset inputs and products are in `data/scenarios/v4_pbh_growth.json` and `results/v4/`. Pinned dependencies and reproduction commands are in `requirements/core.txt` and `docs/guides/reproducibility.md`.

Appendix A Source records and selection checks

A.1 Source versions and measurement alternatives

The catalogue includes CAPERS-LRD-z9 (Taylor et al. 2025b), CANUCS-LRD-z8.6 (Tripodi et al. 2025), J1007_AGN (Schindler et al. 2025), MACS0416-Y1 (Takechi et al. 2026), and SAPPHIRES-ID-8760 (Fudamoto et al. 2025). The first four have Balmer-line mass estimates. MACS0416-Y1 enters the expanded sample because its

broad-line detection is tentative; the D2 component is treated as one nucleus in one host. Its adopted mass and that of CANUCS-LRD-z8.6 are already corrected for lensing. SAPPHIRES-ID-8760 has no published black-hole mass and is excluded from numerical growth calculations; weak Hβ emission and an unidentified foreground contaminant limit its AGN interpretation. The revised CANUCS analysis by Morishita et al. (2026) provides further evidence for the same object and reuses Tripodi's mass.

For J1007_AGN we adopt the observed Hβ mass $(1.152^{+1.011}_{-0.463}) \times 10^8 M_\odot$ while the authors favor a dust-attenuated interpretation. Their correlated dust-corrected alternative is $(4.514^{+5.218}_{-2.320}) \times 10^8 M_\odot$; it raises the reference requirement from 1.113 to 1.222 without adding another object. For CAPERS we adopt the published log mass 7.58 ± 0.15 dex. Its 6.65–8.50 systematic envelope combines different assumptions, including an assumed Eddington luminosity bound, and is not sampled as a measured interval.

Auxiliary coordinates are attributed in the registry to THRILS (Hutchison et al. 2025), UHZ1 spectroscopy (Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue.

A.2 Expanded-sample membership

Seven masses are explicitly flagged as conditional on a broad-line-region interpretation: 6 candidates with intermediate-width Hα emission and possible outflow contamination (Ren et al. 2025), and RUBIES-EGS-49140 (Taylor et al. 2025a), for which an alternative non-AGN interpretation is recorded. These 7 objects enter the expanded sample only.

Four further JADES objects—GS-20057765, GS-20030333, GS-164055, and GN-4685—enter the expanded sample only because their individual broad-Hβ detections are tentative, although broad emission is supported by the stacked spectrum (Juodžbalis et al. 2026a).

GLIMPSE-38548 and GLIMPSE-7404 also enter the expanded sample only; the catalogue classifies them as tentative broad-Hα AGN (Fei et al. 2026). MACS0416-Y1 supplies the seventh tentative detection: its weak broad Hβ component may include outflow emission (Takechi et al. 2026).

A.3 Adopted measurements and uncertain object matches

The catalogue fields `evidence_status` and `evidence_status_basis` record the evidence classification and its source-based rationale. For three pairs of entries, it remains unclear whether both refer to the same object: Baccus GDS_1210_9515 and JADES GS-8083; GS-10013704 and Scholtz 99671; and Scholtz 16745 and 208643. All six entries are excluded from the analysis. For CEERS-2782, we adopt the higher-signal-to-noise RUBIES-EGS-50052 spectrum preferred by the source study; for NX10835, we adopt the newly available NEXUS mass estimate.

A.4 Sensitivity to measurement choice and sample membership

Across the 238 expanded-sample objects, 17 exceed unity at their central masses. Among the 226 objects with reported errors, 13 exceed unity at the 16th percentile, eleven have $P \geq 0.95$, and 19 exceed unity at the 84th percentile.

The adopted Baccus & Xu measurements come from `arXiv:2512.03281v1`. We retain this version as the catalogue baseline to separate measurement revisions from changes in sample membership. The bibliography cites the published article; the measurement records identify the preprint version supplying the adopted values. We test the published measurements below, keeping the object selection fixed to isolate the effect of the measurement changes. Incorporating the published sample in full would also require reviewing its membership and classifications. We matched 44 catalogue objects to the published sample using identifiers and sky positions. For this sensitivity test, we replaced their preprint values with the redshifts, black-hole masses, and mass uncertainties reported in Table 1 of Baccus & Xu (2026). The threshold counts and the five highest-ranked objects remain unchanged in both samples (Table 9). Of five unmatched records, one is excluded for uncertain identity; retaining or omitting the other four leaves these results unchanged. Omitting them leaves 219 primary and 234 expanded-sample objects.

Table 9 here.

Excluding the three uncertain-identity records with eligible masses reduces the primary sample from 226 to 223 objects and the expanded sample from 241 to 238 objects, while preserving threshold counts and top-five rankings in every scenario in Table 7, including the reference probability counts. The median required lifetime-average Eddington ratio across primary objects, evaluated using their central masses, changes from 0.57420 to 0.57425; the objects above the reference threshold remain the same.

Seven alternative-measurement comparisons across 6 objects (2 for CEERS-2782) span mass shifts of -0.600 to $+0.949$ dex and shifts in $\overline{f}_{\mathrm{Edd,req}}$ of -0.083 to $+0.121$ (Fig. 4). Both required Eddington ratios remain below unity in every pair. Testing the objects above the threshold requires additional independent mass estimates.

Figure 4 here.

A.5 Growth-product files and parameter grids

In `data/processed/v3/`, `v3_accreting_measurements.csv` contains the measurement records, and `v3_accreting_objects.csv` contains one adopted record per provisional object. Join them using `measurement_id` and `physical_object_id`; `host_system_id` links systems assigned to the same host. The measurement table records the adopted choice in `preferred_measurement_flag` and its reason. Masses are tabulated as $\log_{10}(M_{\mathrm{BH}}/M_{\odot})$, with uncertainties in dex. Redshift is dimensionless; coordinates are in degrees. Empty numeric cells denote missing values, not zeros. Limits and conditional masses have separate flags. The file `results/v3/selected_samples/tables/publication_object_selection.csv` provides `publication_primary_flag` and `publication_exploratory_flag`; the latter selects the expanded sample, including the primary objects. Source versions and

locators remain in the measurement table and `data/source_provenance_registry.csv`.

The growth products summarized in Table 3 use these adopted measurements and sample flags.

Each admitted object has three required-rate entries, for seed masses of 10^2, 10^4, and $10^5\ M_\odot$; two required-seed entries, for average Eddington ratios of 0.3 and 1; and three duty-cycle entries, for active rates of 1, 2, and 3 with no accretion between active phases. These use $z_{\rm seed} = 30$, $\epsilon = 0.1$, and no merger boost. The compatibility table compares 18 settings (three spin-dependent efficiency prescriptions, two merger multipliers, and three average rates) with four seed-mass intervals, producing 72 checks per object (Appendix C). Together, these four table products contain 80 scenario entries per object, or 19,040 across the expanded sample. The entries share adopted masses and assumptions.

Four additional tables in `results/v3/tables/` provide an object summary (`v3_object_results.csv`; 343 records), grouped field-availability counts (`v3_field_availability.csv`; 688 rows), ten growth-assumption comparisons per eligible object (`v3_growth_assumption_comparison.csv`; 2,410 rows), and 54 minimum-seed combinations per eligible object (`v3_minimum_seed_mass.csv`; 13,014 rows). Applying the expanded-sample flags gives 2,380 comparison rows and 12,852 minimum-seed rows. These tables overlap the core products and are counted separately. Both growth tables reuse the reference mass-error draws. The minimum-seed table includes radiation and uses the settings in `data/scenarios/catalogue_growth_comparisons.json`; its central values agree with matching v4 cases, but the random streams differ.

The seed-mass–accretion-rate map evaluates 100 seed masses and 90 average rates for each of three spin cases and two merger multipliers: 54,000 grid points per object. It assumes $z_{\rm seed} = 30$ and the efficiency prescription in Appendix C.1; a contour marks combinations reproducing the adopted mass. The seed-mass–starting-redshift map evaluates 150 masses and 150 starting redshifts at fixed $\epsilon = 0.1$ and no merger boost, giving 22,500 points per object. The two map types therefore contain 76,500 grid evaluations per admitted object. These maps use central mass estimates.

The object-level products are in `results/v3/`; the conservative samples and their sensitivity tables are selected in `results/v3/selected_samples/`. Publication-selection flags must be applied to obtain the admissible counts above from the broader archival release. Appendix E gives the four-target earlier-onset extension: 564 minimum-seed results, 3,102 seed–onset map entries, and 72 control entries per target, with 10,000 mass-error draws per target. This extension is reported separately from the full-sample grid totals.

Appendix B Growth-model derivations and implementation

B.1 Accretion and exponential growth

For electron scattering in ionized hydrogen, the Eddington luminosity, Eddington ratio, and Eddington timescale are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH} c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \tag{4}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M_{\rm BH}} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon} \frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{5}$$

The inflow rate is defined at the radiating flow; earlier mass loss is omitted. Dividing Eq. 5 by $M_{\rm BH}$ and integrating at fixed ϵ gives

$$\ln\!\left(\frac{M_{\rm BH}}{B_{\rm merge} M_{\rm seed}}\right) = \frac{1-\epsilon}{\epsilon} \frac{1}{t_{\rm Edd}} \int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt = \frac{1-\epsilon}{\epsilon} \frac{\overline{f}_{\rm Edd}\, \Delta t}{t_{\rm Edd}}. \tag{6}$$

Exponentiating gives Eq. 2; solving for the average rate, with a floor of zero, gives Eq. 3.

For a constant positive $f_{\rm Edd}$, the e-folding time is

$$t_{\rm efold} = \frac{\epsilon}{1-\epsilon} \frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{7}$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$.

B.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\!\left(\frac{M_{\rm BH}}{M_\odot}\right) = \log_{10}\!\left(\frac{M_{\rm seed}}{M_\odot}\right) + \log_{10} B_{\rm merge} + \frac{1-\epsilon}{\epsilon} \frac{\overline{f}_{\rm Edd}\, \Delta t}{t_{\rm Edd} \ln 10}. \tag{8}$$

We evaluate $B_{\rm merge} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\rm BH}(B_{\rm merge}=2)}{M_{\rm BH}(B_{\rm merge}=1)} = 2, \qquad \Delta \log_{10} M_{\rm BH} = \log_{10} 2 \simeq 0.3010. \tag{9}$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\rm Edd}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{10}$$

Here $M_{\rm seed,req}$ is the seed mass required for the specified average Eddington ratio, radiative efficiency, growth interval, and merger multiplier.

B.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\,\mathrm{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{11}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected.

For the radiation check, we calculate cosmic ages numerically as

$$t(z) = \frac{1}{H_0}\int_0^{1/(1+z)} \frac{x\,dx}{\sqrt{\Omega_r + \Omega_m x + \Omega_\Lambda x^4}}, \tag{12}$$

where x is the scale factor. We retain the same H_0 and Ω_m, adopt $\Omega_r = \Omega_m/(1+3400) \simeq 9.262\times10^{-5}$, and set $\Omega_\Lambda = 1-\Omega_m-\Omega_r$ to preserve flatness. Including radiation raises the reference required ratios by 0.14–0.26%, preserving the central-mass counts of eleven primary and 17 expanded-sample objects above unity. Two boundary cases change: CEERS-00717's sampled 16th percentile rises from 0.99805 to 1.00003, and A2744-QSO1's dynamical-mass requirement rises from 0.9996 to 1.0016. Both shifts are much smaller than the uncertainty intervals from the mass-error draws.

B.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f}_{\rm Edd} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f}_{\rm Edd,req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{13}$$

Here D is the fraction of time spent active, with $f_{\rm burst} > f_{\rm quiet} \geq 0$. The expression applies when $\overline{f}_{\rm Edd,req} \geq f_{\rm quiet}$. Values $D_{\rm req} > 1$ identify histories that cannot reach the observed mass at the assumed rates.

B.5 Mass-uncertainty implementation

Sampling uses random seed 20260808 and a stable random stream for each object identifier. With 10,000 draws, the Monte Carlo standard error of a probability is at most 0.005, or about 0.0022 at a probability of 0.95; this reflects the finite number of

draws. The 12 NEXUS objects without reported mass errors are excluded from probability calculations and uncertainty rankings. Table 4 distinguishes separate calibration scales from scatter already included in the quoted errors.

For the J1007_AGN calibration convolution, doubling the number of integration points changes the probabilities by less than 10^{-13}. The mass-scale tests add the same fixed offset to every original mass draw. The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M_{\rm BH}/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

B.6 Illustrative growth tracks

Figure 5 shows tracks at four fixed efficiencies, highlighting the five primary objects retaining $P \geq 0.95$ after a -0.5 dex mass shift and GN-z11 as an ultraviolet-mass comparison. Accretion rates are selected separately in each panel for proximity to the observed masses; the panels illustrate possible histories rather than isolating the effect of efficiency. The efficiency boundaries in Figure 3 provide the corresponding comparison at a common required-rate threshold.

Figure 5 here.

For these tracks, we test average Eddington ratios from 0.1 to 2.0 in steps of 0.1. At each seed mass and fixed radiative efficiency, we plot the 2 rates with the most primary objects within 0.5 dex of either merger prediction ($B_{\rm merge} = 1$ or 2), requiring at least 5 objects. Ties favour smaller median absolute residuals, then smaller rates. Most selected objects lie at $z \simeq 4$–6. This selection is for display; conclusions use the required-rate ranking. Reference rankings and seed-mass–redshift maps use $\epsilon = 0.1$.

Appendix C Compatibility across seed masses and accretion rates

C.1 Efficiency prescription

The maps use the ideal Kerr thin-disc radiative efficiency at the innermost stable circular orbit (ISCO; Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3 r_{\rm ISCO}}}, \tag{14}$$

$$r_{\rm ISCO} = 3 + Z_2 - {\rm sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{15}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1 + a)^{1/3} + (1 - a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{16}$$

Here $a = cJ/(GM_{\rm BH}^2)$ is the dimensionless spin, J is the signed angular momentum relative to the disc, and $r_{\rm ISCO}$ is measured in units of $GM_{\rm BH}/c^2$. Neglecting photon capture and stress at the ISCO gives $\epsilon_a = 0.03775$, 0.05719, and 0.42265 for $a = -1, 0, +1$. For supercritical inflow, we adopt a logarithmic luminosity relation motivated by advection and wind models (Poutanen et al. 2007), with coefficient 1

and no radial mass loss:

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \le 1, \\ \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{17}$$

The logarithmic relation applies only for $f_{\rm Edd} > 1$. Each cell in Figure 6 holds spin, Eddington ratio, and effective radiative efficiency constant, so $f_{\rm Edd} = \overline{f}_{\rm Edd}$; relativistic slim-disc dynamics are omitted. If efficiency varies, growth depends on $\int [(1 - \epsilon(t))/\epsilon(t)] f_{\rm Edd}(t)\, dt$ and requires both histories.

Figure 6 here.

C.2 Seed-mass ranges and compatible fractions

We solve Eq. 10 for each object's required seed mass and check whether it lies within each of four intervals, including the endpoints: 10–100 $M_\odot$ (light), 10^3–10^4 $M_\odot$ (intermediate), 10^4–10^6 $M_\odot$ (heavy), and 10^2–10^6 $M_\odot$ (broad). The grid uses $z_{\rm seed} = 30$, spins $a = -1, 0, +1$, merger multipliers $B_{\rm merge} = 1, 2$, and average Eddington ratios 0.3, 1, and 2, with the efficiency prescription in Section C.1.

The intervals overlap and carry no probability weights; the broad interval covers several possible seed-formation channels. A required seed below an interval's lower bound is incompatible: even its smallest seed would grow beyond the observed mass under those assumptions. We report the percentage of objects whose required seeds fall within each interval, using central mass estimates without propagating their errors. These percentages describe the literature sample. Population fractions would require the contributing studies' selection and detection probabilities.

The two samples show similar patterns (Fig. 6). For heavy seeds at $a = +1$ and $f_{\rm Edd} = 1$, changing $B_{\rm merge}$ from 1 to 2 raises compatibility from 57 to 70 per cent in the primary sample and from 58 to 70 per cent in the expanded sample. Higher accretion rates can reduce compatibility when even the smallest seed grows beyond the observed mass.

Appendix D Seed mass and starting redshift

Figure 7 extends the two starting redshifts in Figure 3 to $11 \le z_{\rm seed} \le 30$ for UNCOVER-20466 and GN-z11. Later growth onset increases the required average Eddington ratio; larger seeds reduce it. Maps for all objects are available in the repository under `results/v3/parameter_maps/seedredshift_mass_maps/`.

Figure 7 here.

Appendix E Earlier accretion onset for four targets

We compare earlier accretion onset for CAPERS-LRD-z9, UNCOVER-20466, and CANUCS-LRD-z8.6, the three highest-ranked primary objects under the reference growth model, together with GN-z11 as an expanded-sample UV-mass comparison object. We use the growth component of Eq. (1) and Section 2 of Dayal (2024), with an optimistic accretion onset at matter–radiation equality ($z_{\rm acc} = 3400$), and separately test delayed onset. The seed mass denotes the mass at the start of accretion. For every

onset, including the $z_{\rm acc} = 30$ controls, we use the radiation-inclusive cosmic ages in Eq. 12.

With no mergers and constant radiative efficiency, the minimum onset seed that reaches an adopted mass for $0 \leq \overline{f}_{\rm Edd} \leq f_{\rm cap}$ is

$$\log_{10}\left(\frac{M_{\rm seed,min}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - f_{\rm cap}\frac{1-\epsilon}{\epsilon}\frac{t(z_{\rm obs}) - t(z_{\rm acc})}{t_{\rm Edd}\ln 10}. \tag{18}$$

Here $f_{\rm cap}$ is a cap on the lifetime-average luminosity-based Eddington ratio. We evaluate 47 onset redshifts between 20 and 3400, caps of 0.3, 1, and 2, and efficiencies of 0.1 and $1 - \sqrt{8/9} = 0.05719$. Efficiency remains fixed at every cap. Reported asymmetric log-mass errors are propagated with 10,000 equal-side half-normal Monte Carlo draws per object, using stable object-specific streams with seed 20261001. Redshifts remain fixed; a -0.5-dex mass shift is tested separately on the same draws. The onset grid is recorded in the analysis configuration listed under Data and code availability. Table 10 here.

At $\epsilon = 0.1$ and $f_{\rm cap} = 1$, equality-onset minimum seeds are approximately 1144, 1184, 1018, and 267 $M_\odot$, respectively (Table 10). Thus an earlier start alone does not allow a 100 $M_\odot$ seed to reach any of these four central masses under these assumptions. Figure 8 here. Relative to $z_{\rm acc} = 30$, equality onset adds 98.73 Myr and reduces each minimum seed by a factor of 7.20. Moving onset from $z_{\rm acc} = 100$ to 3400 adds only about 16.35 Myr and reduces it by a further factor of 1.39. Lower efficiency or larger allowed average rates reduce the required seeds further. The lower-efficiency control permits 100 $M_\odot$ seeds starting at $z = 30$ to reach all four central masses.

E.1 Interpreting primordial-seed growth limits

Equal masses at accretion onset and equal growth parameters give identical growth regardless of seed origin. The $z = 30$ starting time and equality-onset accretion are assumed benchmarks. A primordial black hole (PBH) interpretation requires a seed present before equality; its formation, pre-onset growth, survival, gas supply, abundance, and external observational constraints remain outside this calculation. Section 3 of Dayal (2024) additionally models halo assembly, gas, star formation, and feedback for UHZ1 and GHZ9. Dayal (2026) compares astrophysical and primordial seeding using models that include gas supply, star formation, feedback, and population predictions. The four-target analysis here isolates the effects of seed mass, accretion onset, and radiative efficiency on the required growth.

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Author contributions

Nicole Jiang is the sole author and is responsible for the research and manuscript.

Figures and legends

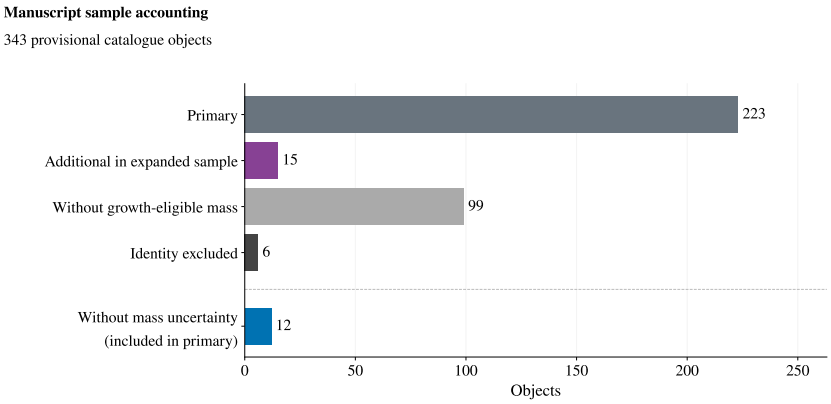


Fig. 1 Sample accounting for 343 provisional catalogue objects. The first four bars show mutually exclusive categories: 223 primary objects, 15 additional expanded-sample objects, 99 retained without eligible masses, and 6 identity exclusions. The final bar shows the 12 primary objects without reported mass uncertainties; these are included in the primary count. The expanded sample contains 238 objects. Probability calculations use 211 primary and 226 expanded-sample objects.

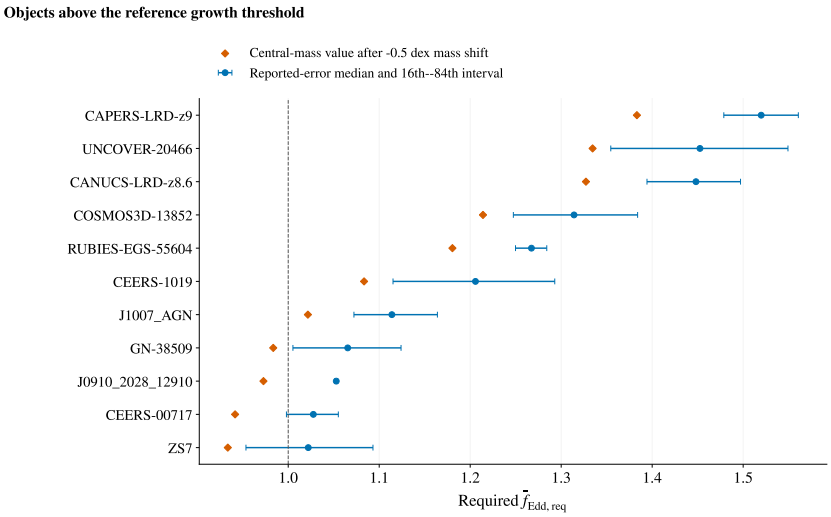


Fig. 2 The eleven primary objects with $f_0 > 1$, where f_0 is the required lifetime-average Eddington ratio at the adopted central mass under the reference model. Circles and bars show medians and 16th–84th percentile intervals of the required lifetime-average Eddington ratio over each object's mass-error draws; diamonds show f_0 after a -0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

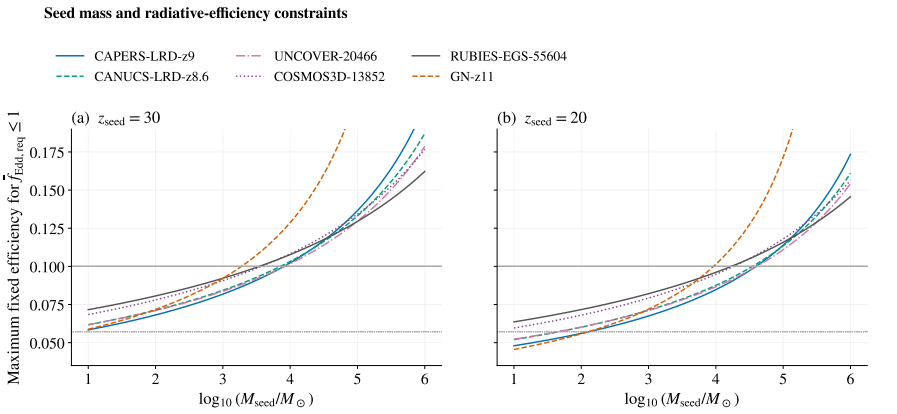


Fig. 3 Seed-mass–efficiency boundaries for Eddington-limited average growth at 2 starting redshifts. Each curve uses an adopted central mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and 0.05719 (dotted grey). The targets are the five primary objects retaining $P \geq 0.95$ after a -0.5 dex mass shift under the reported-error model, and GN-z11 from the expanded sample. Each point assumes a radiative efficiency that remains constant throughout growth.

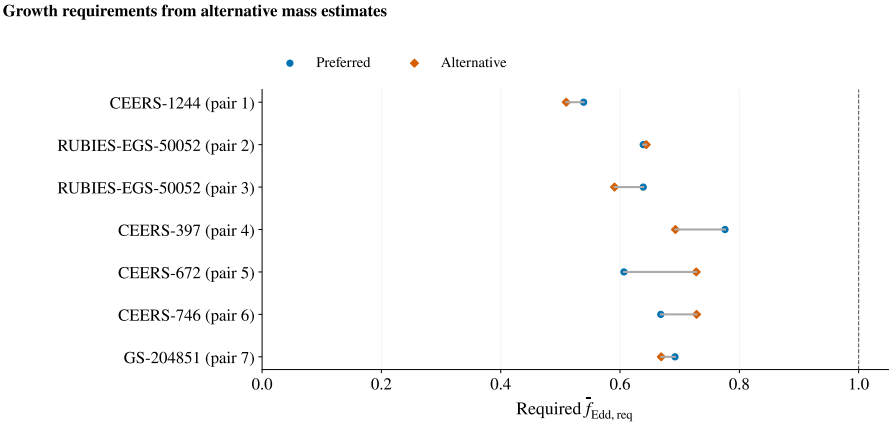


Fig. 4 Preferred (circles) and alternative (diamonds) required ratios at the central masses for 7 measurement pairs across 6 primary objects. Pair numbers follow the released alternative-measurement table, which supplies stable measurement identifiers; RUBIES-EGS-50052 (CEERS-2782) has 2 alternatives. All pairs remain below the reference threshold.

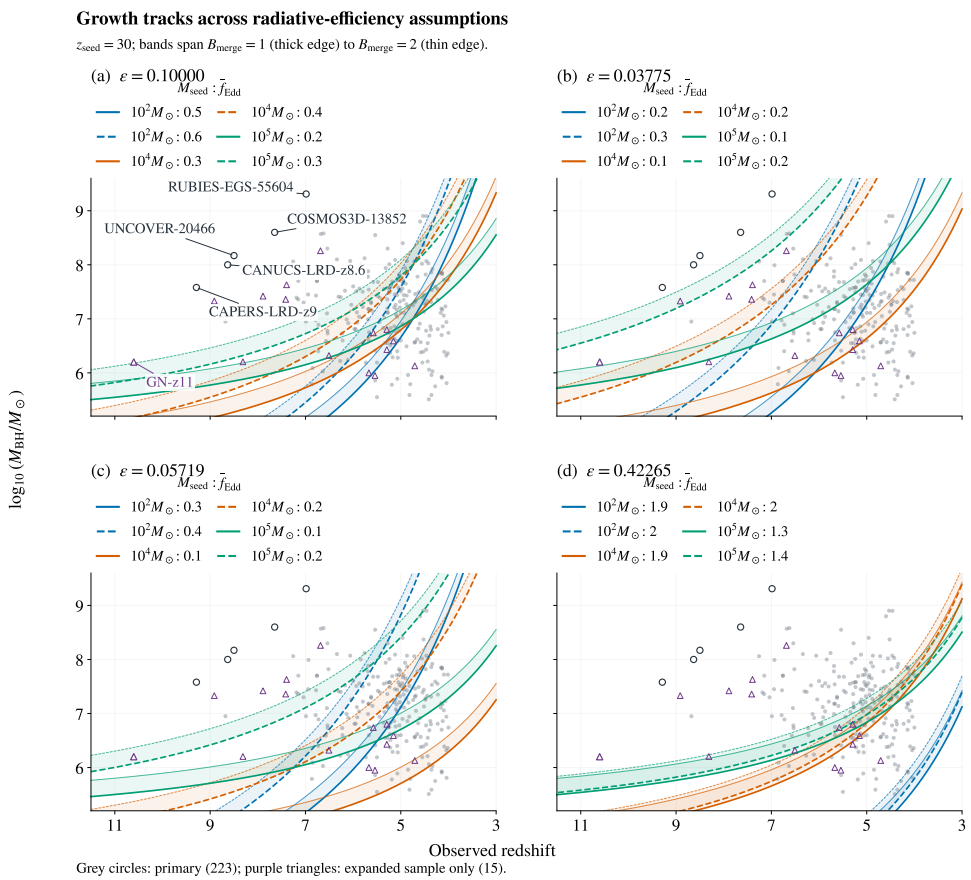


Fig. 5 Growth tracks with 223 primary and 15 additional objects. Panels show 4 constant radiative efficiencies at $z_{\rm seed} = 30$; colours distinguish seeds of 10^2, 10^4, and $10^5 M_\odot$, and legends list the average Eddington ratios, selected separately in each panel to lie near the sample. Shaded bands span merger boosts $B_{\rm merge} = 1$–2, with thick and thin edges respectively. The five primary objects retaining $P \geq 0.95$ after a -0.5 dex mass shift under the reported-error model, together with GN-z11, are labelled in the first panel and highlighted in every panel. Objects with unresolved identities or without eligible masses are omitted.

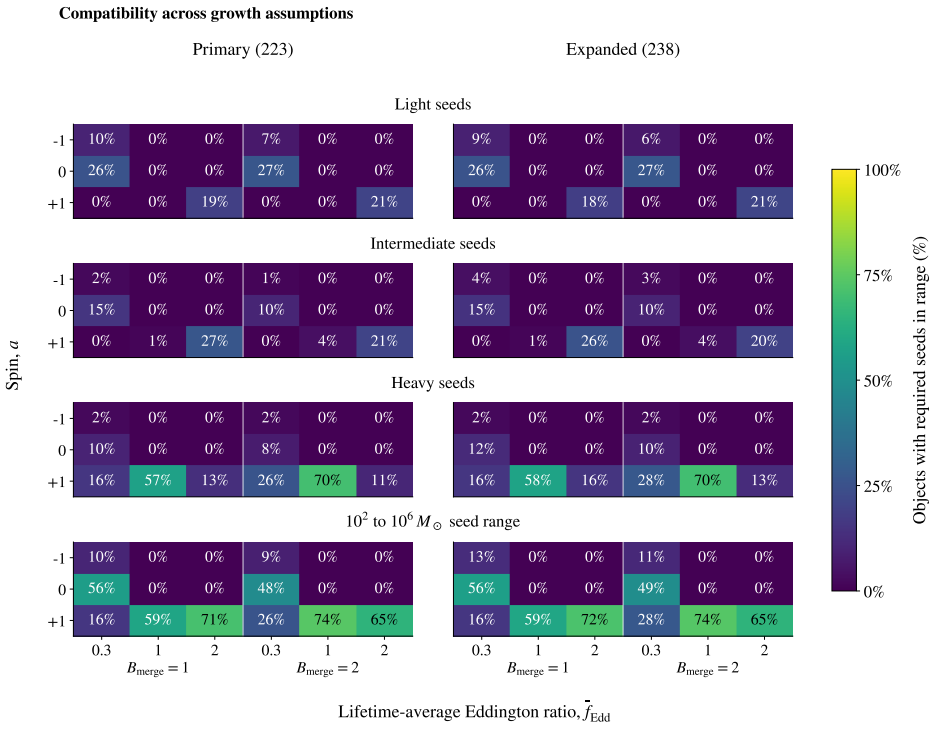


Fig. 6 Constant-state compatibility fractions in the primary sample (left) and the expanded sample (right). The expanded sample includes all primary objects. Each column specifies merger multiplier B_{merge} and fixed luminosity ratio f_{Edd}; rows specify ideal spin-dependent efficiency, with the super-critical prescription in Section C.1. Seed intervals overlap and have no assigned probability weights.

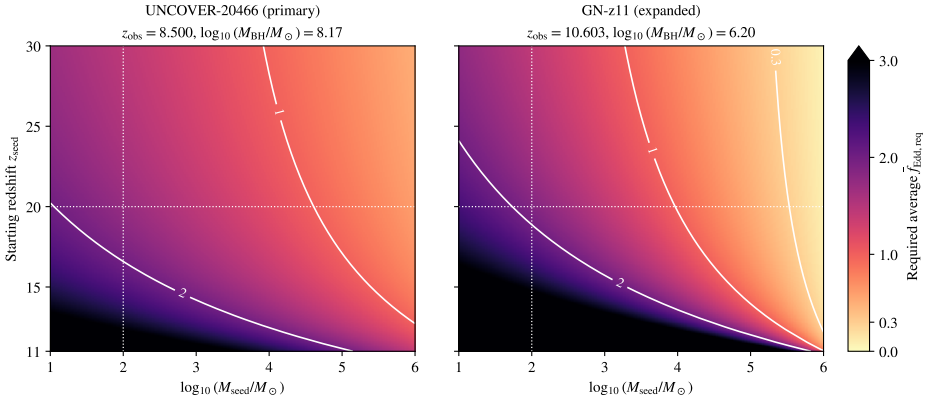


Fig. 7 Required average Eddington ratio for UNCOVER-20466 (primary sample) and GN-z11 (expanded sample), using their adopted central masses and redshifts shown above each panel. Both panels use $\epsilon = 0.1$, no merger boost, and the same axes and colour scale. White contours mark required ratios of 0.3, 1, and 2; dotted guides mark a $100\,M_\odot$ seed and $z_{\rm seed} = 20$. Colours saturate above a required ratio of 3. Mass uncertainties are not propagated in these maps.

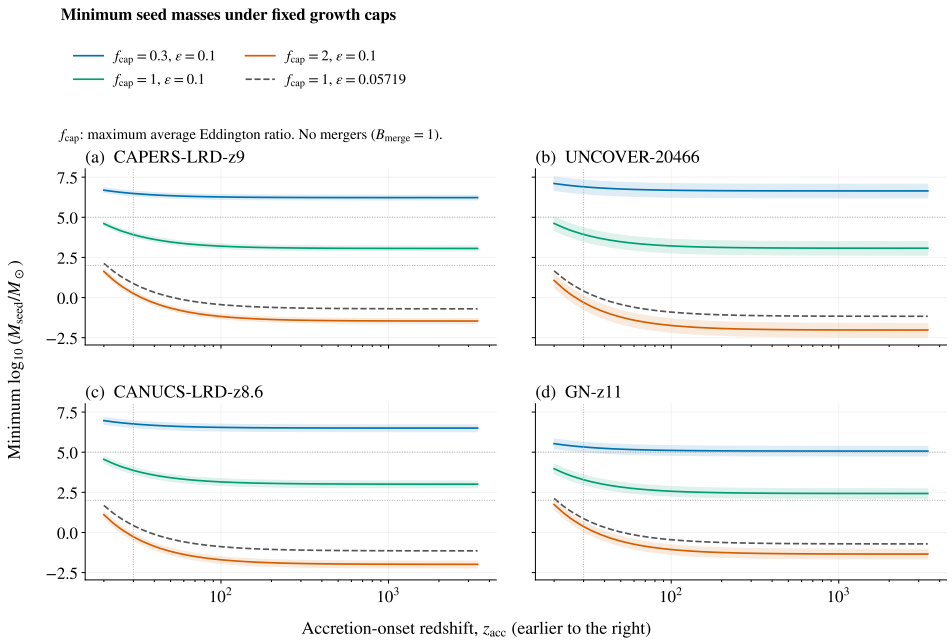


Fig. 8 Minimum onset seed mass versus accretion-onset redshift for the four targets. Solid curves vary the average Eddington-ratio cap at fixed $\epsilon = 0.1$; shaded bands propagate the 16th–84th percentiles of reported mass errors. Dashed curves show the lower-efficiency sensitivity. Earlier onset reduces the required mass, with diminishing gains in available time.

Tables

Table 1 Catalogue contents and analysis samples. Measurements are literature records of individual objects; an object may have records from several studies. Object counts are provisional where identifications across studies remain uncertain.

Category	Count
Measurements / provisional objects / host systems	355 / 343 / 342
Growth-eligible measurements / objects	248 / 241
Preliminary Balmer selection: measurements / objects	237 / 230
Primary sample / expanded sample	223 / 238
Objects with mass uncertainties (primary / expanded)	211 / 226
Objects excluded for uncertain identity	6
Objects without growth-eligible measurements	102

The preliminary Balmer selection requires a non-conditional Balmer virial mass and secure or probable evidence for an accreting black hole in the catalogue. These counts precede the final checks of individual line detections and object identities. Of these 230 objects, 4 tentative JADES detections enter only the expanded sample and 3 are excluded for uncertain identity, leaving 223 primary objects. The 102 objects without eligible masses include 3 of the 6 identity exclusions. Figure 1 counts those 3 under identity exclusions, leaving 99 in its bar for objects without eligible masses.

Table 2: Source-family inventory. Counts are catalogue measurements / growth-eligible measurements / primary-sample objects (M/E/P). The columns sum to 355/248/223; E includes alternatives, whereas P counts each selected object once. "Balmer" denotes single-epoch virial masses. "Context" denotes mass estimates retained for discussion but excluded from growth calculations because of their method or interpretation. "Upper limit" denotes an upper bound rather than a measured central mass; these bounds are also excluded from growth calculations.

Source	M/E/P	Mass	Selection channel; principal caveat
Baccus Baccus & Xu (2026)	49/49/48	Balmer	Frozen preprint masses; calibration
Chavez Ortiz Chávez Ortiz et al. (2026)	1/0/0	Context	Model-dependent mass
Chisholm Chisholm et al. (2024)	1/0/0	—	No canonical mass
Davis / THRILS Davis et al. (2026)	6/6/6	Balmer	Virial calibration
Fei / GLIMPSE Fei et al. (2026)	10/10/8	Balmer	Lensing corrections
Greene / UNCOVER Greene et al. (2024)	9/9/9	Balmer	Lens model; broad-line-region interpretation
Harikane Harikane et al. (2023)	10/10/5	Balmer	Local virial calibration
Juodzbalis / JADES Juodžbalis et al. (2026a)	23/23/17	Balmer	Extinction; calibration
Killi Killi et al. (2024)	1/1/1	Balmer	Linear mass-error conversion
Kocevski / LRDs Kocevski et al. (2025)	6/6/6	Balmer	Dust correction; calibration
Larson / CEERS-1019 Larson et al. (2023)	1/1/1	Balmer	Broad-component significance
Lin / ASPIRE Lin et al. (2024)	16/16/16	Balmer	Virial calibration
Lin / COSMOS-3D Lin et al. (2026)	13/13/13	Balmer	Possible scattering bias
Lyu / SMILES Lyu et al. (2024)	19/0/0	—	Spectral-energy-distribution AGN identification

Table 2 continued

Source	M/E/P	Mass	Selection channel; principal caveat
Maiolino / GN-z11 Maiolino et al. (2024)	1/1/0	UV	Ultraviolet estimator outside primary
Mascia Mascia et al. (2026)	8/0/0	—	No canonical mass
Matthee / EIGER-FRESCO Matthee et al. (2024)	20/20/19	Balmer	Dust; virial calibration
Mazzolari Mazzolari et al. (2025)	25/0/0	—	Star-formation overlap
Leung / MEOW Leung et al. (2026)	12/0/0	—	Spectral-energy-distribution AGN fraction
Naidu / MoM-BH*-1 Naidu et al. (2026)	1/0/0	Context	Scattering; stellar alternative
Napolitano / GHZ9 Napolitano et al. (2025a)	1/0/0	Context	Mass assumes Eddington rate
Napolitano / GHZ4,7 Napolitano et al. (2025b)	2/0/0	—	Tentative diagnostics
Ren / ALPINE-CRISTAL Ren et al. (2025)	7/7/1	Balmer	Conditional broad-line-region interpretations
Scholtz / JADES Scholtz et al. (2025)	21/0/0	—	Tentative classifications
Skyfire / CEERS Kocevski et al. (2026)	22/22/22	Balmer	Virial calibration
Tang Tang et al. (2025)	3/0/0	—	Ionizing-source ambiguity
Taylor / CEERS-RUBIES Taylor et al. (2025a)	37/37/35	Balmer	Slit loss; dust; calibration
Treiber / UNCOVER Treiber et al. (2025)	2/0/0	—	Stellar alternatives
Ubler / ZS7 Übler et al. (2024)	1/1/1	Balmer	Error includes calibration scatter
Bogdan / Zou / UHZ1 Bogdán et al. (2024); Zou et al. (2026)	2/0/0	Context	Later detection disputed
Zhang / LRDs Zhang et al. (2026)	5/0/0	Upper limit	Mass upper limits
Zhuang / NEXUS Zhuang et al. (2026)	15/12/12	Balmer	Twelve masses lack errors
Taylor / CAPERS Taylor et al. (2025b)	1/1/1	Balmer	Prism resolution; uncertain broad-line geometry and virial calibration
Tripodi / CANUCS Tripodi et al. (2025)	1/1/1	Balmer	Lensing model; uncertain extinction treatment and virial calibration
Schindler / J1007 Schindler et al. (2025)	1/1/1	Balmer	Total Hβ luminosity estimator; correlated dust correction and calibration scatter
Takechi / Y1 Takechi et al. (2026)	1/1/0	Balmer	Tentative broad Hβ; possible outflow contamination
Fudamoto / SAPPHIRES Fudamoto et al. (2025)	1/0/0	—	No published black-hole mass; weak Hβ and foreground contamination

Table 3 Core growth-testing coverage for the admissible expanded sample. Counts exclude objects without eligible masses, unresolved-identity exclusions, and status-only records. Per-object counts refer to the same adopted mass under different assumptions; grid points are numerical evaluations, not independent physical models.

Product	Per object	Sample total
Required accretion rates	3	714
Required seed masses	2	476
Duty-cycle requirements	3	714
Growth compatibility checks	72	17,136
Four core tabulated products	80	19,040
Numerical parameter maps	2	476
Seed–accretion grid evaluations	54,000	12,852,000
Seed–onset grid evaluations	22,500	5,355,000
Combined grid evaluations	76,500	18,207,000
Monte Carlo mass draws[a]	10,000	2,260,000

The expanded sample has 238 objects, including 223 primary objects. The additional summary, availability, comparison, and minimum-seed tables in Section 3.4 are counted separately. [a] Only the 226 expanded-sample objects with reported mass errors are sampled; the other 12 enter central-mass calculations only. The primary sample alone contributes 17,059,500 map-grid evaluations and 2,110,000 mass draws for its 211 objects with errors. The archival release additionally retains numerical entries for 3 identity-excluded objects and status records for 102 objects without eligible masses; none contributes to the counts here. The four-target earlier-onset extension has 2,256 minimum-seed results, 12,408 seed–onset map entries, and 288 control entries, reported separately (Appendix E).

Table 4 The eleven primary objects above the reference threshold, ordered by f_0. Redshifts are the adopted spectroscopic values. Masses and errors are in $\log_{10}(M_{\rm BH}/M_\odot)$; all estimators are single-epoch virial. f_0 is the required lifetime-average Eddington ratio evaluated at the adopted central mass under the reference model; f_{16} is the 16th percentile of this required ratio over the object's mass-error draws, and $P_0 = P(\overline{f}_{\rm Edd,req} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Δm (dex) is the reduction in log mass that brings the central required average Eddington ratio to 1, with the growth assumptions unchanged. All probabilities use the quoted mass errors and fixed growth assumptions. Additional mass-estimator limitations are in Table 8.

Object/source	z	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-	Δm
CAPERS-LRD-z9	9.288	Hβ	$7.58^{+0.15}_{-0.15}$	1.519	1.479	> 0.999	1.383	> 0.999	1.907
Taylor et al. (2025b)									
UNCOVER-20466	8.5	Hβ	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999	1.921
Greene et al. (2024)									
CANUCS-LRD-z8.6	8.6319	Hβ	$8.00^{+0.20}_{-0.22}$	1.448	1.394	> 0.999	1.327	> 0.999	1.856
Tripodi et al. (2025)									
COSMOS3D-13852	7.646	Hβ	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999	1.576
Lin et al. (2026)									
RUBIES-EGS-55604[a]	6.986	Hβ	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999	1.541
Kocevski et al. (2025)									
CEERS-1019	8.679	Hβ	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822	0.842
Larson et al. (2023)									
J1007_AGN[d]	7.2583	Hβ	$8.06^{+0.27}_{-0.22}$	1.113	1.072	0.997	1.022	0.702	0.618
Schindler et al. (2025)									
GN-38509[b]	6.678	Hα	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392	0.398
Juodžbalis et al. (2026a)									
J0910_2028_12910[a]	6.6214	Hα	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001	0.330
Baccus & Xu (2026)									
CEERS-00717	6.936	Hα	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018	0.160
Harikane et al. (2023)									
ZS7[c]	7.15	Hβ	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173	0.130
Übler et al. (2024)									

[a] A separate 0.5-dex calibration scale is reported for RUBIES-EGS-55604 and J0910_2028_12910; these scales are omitted from the quoted errors and probabilities shown here. [b] GN-38509's quoted mass errors already include 0.3-dex calibration scatter. [c] ZS7's quoted 0.4-dex error already includes calibration scatter. [d] J1007_AGN has a separate 0.43-dex calibration scale, omitted here and tested in Table 6. Unmarked rows have no separate scalar numerical calibration scale recorded; their quoted errors may omit other mass-estimator uncertainties.

Table 5 Sensitivity to fixed mass offsets under the reference model. The slash-separated counts are for objects above unity at the central mass, at the 16th percentile, and with conditional $P(\overline{f}_{\rm Edd,req} > 1) \geq 0.95$. The probability-based counts use 211 primary and 226 expanded-sample objects with reported errors; central-mass counts use all 223 and 238 objects.

Mass offset (dex)	Primary	Expanded
-0.5	7 / 5 / 5	10 / 7 / 6
-0.3	9 / 8 / 6	13 / 10 / 7
0	11 / 9 / 8	17 / 13 / 11
$+0.3$	14 / 14 / 13	21 / 18 / 17
$+0.5$	18 / 15 / 14	25 / 21 / 18

Table 6 Conditional probabilities of exceeding the reference required average Eddington ratio of 1. Reported-error probabilities use the adopted error models without Monte Carlo sampling. The calibration columns add independent Gaussian scatter: 0.5 dex for RUBIES-EGS-55604 and J0910_2028_12910, and 0.43 dex for J1007_AGN. The symmetric cases use exact normal tails; J1007's asymmetric errors use numerical convolution. The baseline Monte Carlo results and sample membership are unchanged.

	No mass shift		−0.5 dex shift	
Object	Reported	+ calibration	Reported	+ calibration
RUBIES-EGS-55604	> 0.999	0.999	> 0.999	0.979
J0910_2028_12910	> 0.999	0.745	< 0.001	0.367
J1007_AGN	0.997	0.901	0.701	0.607

Table 7 Objects requiring $\overline{f}_{\rm Edd,req} > 1$ at their central masses under fixed growth assumptions. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use central mass estimates and provisional object identities.

Scenario	Primary (223)	Expanded (238)
Reference	11	17
$M_{\rm seed} = 10^3 M_\odot$	5	7
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	21	28
$\epsilon = 0.05719$	0	0

Table 8 Mass-estimator limitations for the 11 primary objects requiring $\overline{f}_{\rm Edd,req} > 1$ under the reference assumptions. Caveats summarize the source records cited in Table 4.

Object	Mass-estimator caveat
CAPERS-LRD-z9	Hβ; dense gas and broad-line-region geometry, correlated dust alternatives; the 0.15 dex error includes calibration coefficients but not the full high-redshift systematic.
CANUCS-LRD-z8.6	Hβ; mass already lens-corrected; alternate lens model and calibration-scatter decomposition remain caveats.
J1007_AGN	Observed total-Hβ luminosity estimator; correlated dust correction raises the mass; separate 0.43 dex calibration scale tested in Table 6.
UNCOVER-20466	Hβ; mass depends on lens model and broad-line-region interpretation.
COSMOS3D-13852	Hβ; possible scattering bias; 1–2 dex shifts tested here.
RUBIES-EGS-55604	Hβ; dust-corrected mass; separate 0.5 dex calibration scale.
CEERS-1019	Weak broad Hβ component.
GN-38509	Hα; quoted mass errors include 0.3 dex calibration scatter.
J0910_2028_12910	Hα; 0.01 dex formal error excludes a 0.5 dex calibration scale.
CEERS-00717	Hα; local calibration; no quantitative systematic uncertainty reported.
ZS7	Hβ; quoted 0.4 dex already includes calibration scatter.

Table 9 Effect of replacing the adopted Baccus preprint values with published measurements, with unresolved identities excluded throughout. Counts denote required ratios above unity at the central masses, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. The adopted catalogue values define the baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Adopted catalogue values	223	11/9/8
Primary	Published; retain unmatched	223	11/9/8
Primary	Published; omit unmatched	219	11/9/8
Expanded	Adopted catalogue values	238	17/13/11
Expanded	Published; retain unmatched	238	17/13/11
Expanded	Published; omit unmatched	234	17/13/11

Table 10 Minimum onset seed masses ($M_\odot$) for the four targets at their adopted central masses, $\epsilon = 0.1$, $f_{\rm cap} = 1$, and no mergers. All columns use radiation-inclusive cosmic ages. Values are rounded; Figure 8 shows the reported-error sensitivity.

Object	$z_{\rm acc} = 30$	$z_{\rm acc} = 100$	$z_{\rm acc} = 3400$
CAPERS-LRD-z9	8243	1587	1144
UNCOVER-20466	8527	1642	1184
CANUCS-LRD-z8.6	7334	1412	1018
GN-z11	1924	370	267